MUHAMMAD ABBAS RANA

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EDUCATION

COMSATS University Islamabad, Lahore Campus Sep 2023 – Jun 2027 (Expected)
Bachelor of Science in Computer Science, **CGPA:** 3.67/4.00

RESEARCH EXPERIENCE

Undergraduate Research Lead (Final-Year Project), COMSATS University Islamabad Sep 2026 – Present
Thesis: Multi-Task Hate Speech, Offensive Language, and Abusive Content Detection in Urdu
Supervisors: Dr. M. Aksam Iftikhar, Dr. M. Shahid Bhatti

- Direct a 3-member research team to curate and preprocess an expanding 214,000+ comment corpus of native Urdu, Roman Urdu, and code-switched social media text for multilingual Transformer models.
- Formulated a multi-layer annotation taxonomy incorporating a dedicated stance layer for quoted hate and counter-speech, refining guidelines across iterative pilot rounds to ensure inter-annotator agreement.
- Develop a functional and behavioral test suite for Urdu and Roman Urdu hate speech detection models to address critical evaluation gaps in low-resource South Asian NLP literature.
- Engineer a browser extension for real-time social media moderation, enabling configurable content blurring across hate, abusive, and profane text with granular user controls for target categories and severity levels.

WORK EXPERIENCE

AI & ML Engineering Intern, DigiSync Technologies Private Limited Jul 2026 – Oct 2026

- Trained and tuned classification models (Random Forest, Gradient Boosting, Logistic Regression) via scikit-learn pipelines and GridSearchCV, evaluating across Stratified k-Fold splits using ROC-AUC and Macro-F1.
- Built a RESTful FastAPI service with async routing, Pydantic validation, and pytest suites, automating containerized CI/CD delivery to AWS EC2 via Docker and GitHub Actions (Ruff, CodeQL, SSH).
- Configured and hardened the production Linux server with an Nginx HTTPS reverse proxy, UFW security rules, systemd service units, and automated cron health monitoring.

PROJECTS

PolicyJEV: Evaluating a System One Model for Toxic-Comment Moderation Sep 2026 – Oct 2026
Tech Stack: Python, scikit-learn, PyTorch, FastAPI, React, TypeScript

- Benchmarked TypeSafe's JEV model in a zero-shot setting against TF-IDF Logistic Regression and fine-tuned DistilBERT on toxic-comment classification (Jigsaw, 63,978 held-out comments) under an evaluation protocol frozen in Git prior to test scoring.
- Demonstrated that while JEV trailed both trained baselines in mAP (0.527 vs. 0.583 and 0.677), it achieved the highest recall on rare toxicity labels under strict human-review budgets (1–2%).
- Engineered a side-by-side interactive comparison console to inspect model moderation decisions and archived the reproducibility package on Zenodo.

Real-Estate Price Analysis: Buenos Aires and Mexico Aug 2026 – Oct 2026
Tech Stack: Python, scikit-learn, pandas, Seaborn

- Trained linear regression models (OLS, Ridge, Lasso) to predict Buenos Aires apartment prices from area and geolocation, reducing test RMSE by 42.6% ($33,070 vs. $57,576 mean baseline across 1,250 held-out listings).
- Performed variance decomposition across 1,322 Mexican property listings in 21 states, demonstrating that 70.7% of price-per-m^2 variance occurs within states and 29.3% between states.

PortalPro: Role-Based Desktop Management System May 2024 – Jun 2024
Tech Stack: Java, JavaFX, MySQL, JDBC

- Led a 3-member team to develop a desktop application with Admin, Student, and Teacher portals; implemented the Java backend and designed the normalized MySQL schema and UML class diagram.

TEACHING EXPERIENCE

Online Computer Science Educator, Eximus Education Dec 2025 – Present

- Instruct AQA A-Level Computer Science students in algorithms, data structures, and programming, guiding a 6-student cohort to 100% A*/A attainment (5 A*, 1 A).
- Design lesson plans, pseudocode walkthroughs, and diagnostic exercises tailored to UK examination board standards to address individual knowledge gaps and strengthen algorithmic problem-solving.

Computer Science & IT Tutor, Self-Employed Jan 2024 – Present

- Mentor students in Cambridge IGCSE Computer Science and A-Level Information Technology via one-to-one conceptual coaching and exam preparation, achieving 100% A* attainment (4 IGCSE CS, 1 A-Level IT).

SKILLS

- **Programming Languages:** Python, C++, Java, SQL, Bash
- **ML & NLP:** PyTorch, TensorFlow, Hugging Face (XLM-RoBERTa, MuRIL, DistilBERT), scikit-learn
- **Data Science & Databases:** NumPy, pandas, Matplotlib, Plotly, PostgreSQL, MySQL, Excel (Power Query)
- **Backend, Cloud & DevOps**: FastAPI, Docker, AWS (EC2), Nginx, GitHub Actions (CI/CD), Linux, Git

HONORS AND AWARDS

Certificate of Distinction, Robotics Program (2021–2023), Punjab Group of Colleges Aug 2023
Third Place (RC Car), PGC Robotics Exhibition, Punjab Group of Colleges Jul 2023
Gold Medal (RC Car), PGC Robotics Exhibition, Punjab Group of Colleges Aug 2022

- Led a 5-member team to build and program an Arduino-based RC vehicle, securing 1st place overall across 18 participating campuses.

CERTIFICATIONS

Data Science Modeling, WorldQuant University Oct 2026
Machine Learning Specialization, DeepLearning.AI & Stanford Online (Coursera) Sep 2026

LEADERSHIP AND VOLUNTEER EXPERIENCE

Volunteer Fundraiser, The Citizens Foundation Sep 2024 – Dec 2024

- Raised donations and conducted community outreach within a 25+ member team for one of Pakistan's largest non-profit organizations, supporting quality education for less privileged children across 2,000+ schools.

Youth Delegate, Pakistan Youth Leadership & Dialogue Camp (Seeds of Peace) Jul 2021

- Selected as 1 of 22 delegates nationwide through a multi-stage screening process for a 5-day residential program focused on cross-cultural dialogue and team-based leadership challenges.